1. Data Sources

Clinical history
NHANES 2007–2012
n = 6,329 children

Spirometry
NHANES 2007–2012
92.3% available

Lung sound
SPRSound (CC BY 4.0)
747 children · 5,216 clips

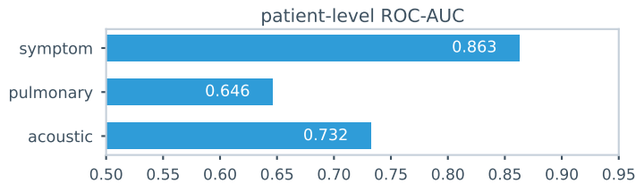
Agents are trained on separate cohorts; no jointly-labelled multimodal dataset is required.

2. Modality Agents

Symptom Agent
calibrated logistic regression

Pulmonary Agent
calibrated logistic regression

Acoustic Agent
AST embeddings + 58 features



3. Evidence Bundle

calibrated probability

named contributing factors

modality-present flag

Each agent emits the same structure, so a missing modality drops a bundle without changing the interface.

4. Decision Agent

Logistic meta-learner
over agent evidence

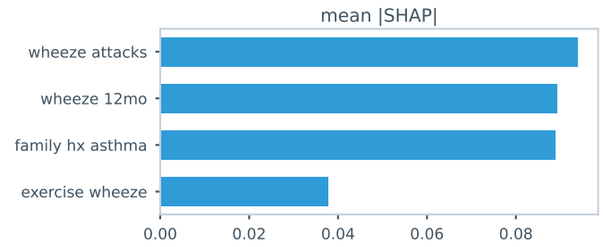
Abstains on low confidence
flags inter-agent conflict

fused ROC-AUC 0.871
0.880 at 85% coverage

Refers the 15% of children closest to the operating threshold for clinical review.

5. Explanation Fusion

Unified rationale across modalities



deletion-verified: 2.8× random
sufficiency 0.017 vs 0.084

6. Clinical Output

Asthma likely

Asthma unlikely

Refer for review

